

Excel in SAS®

An End-to-End PROC SQL Workflow

Minnesota SAS Users Group

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SAS Education



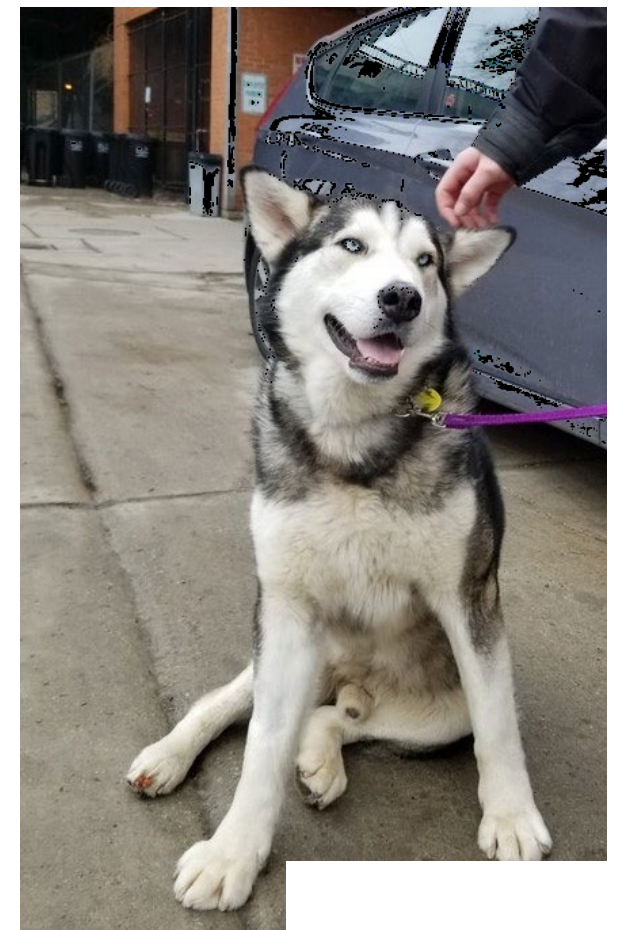
Bio

As a SAS Instructor, I draw on my computer systems background, using logic, visuals, and analogies to spark critical thinking.

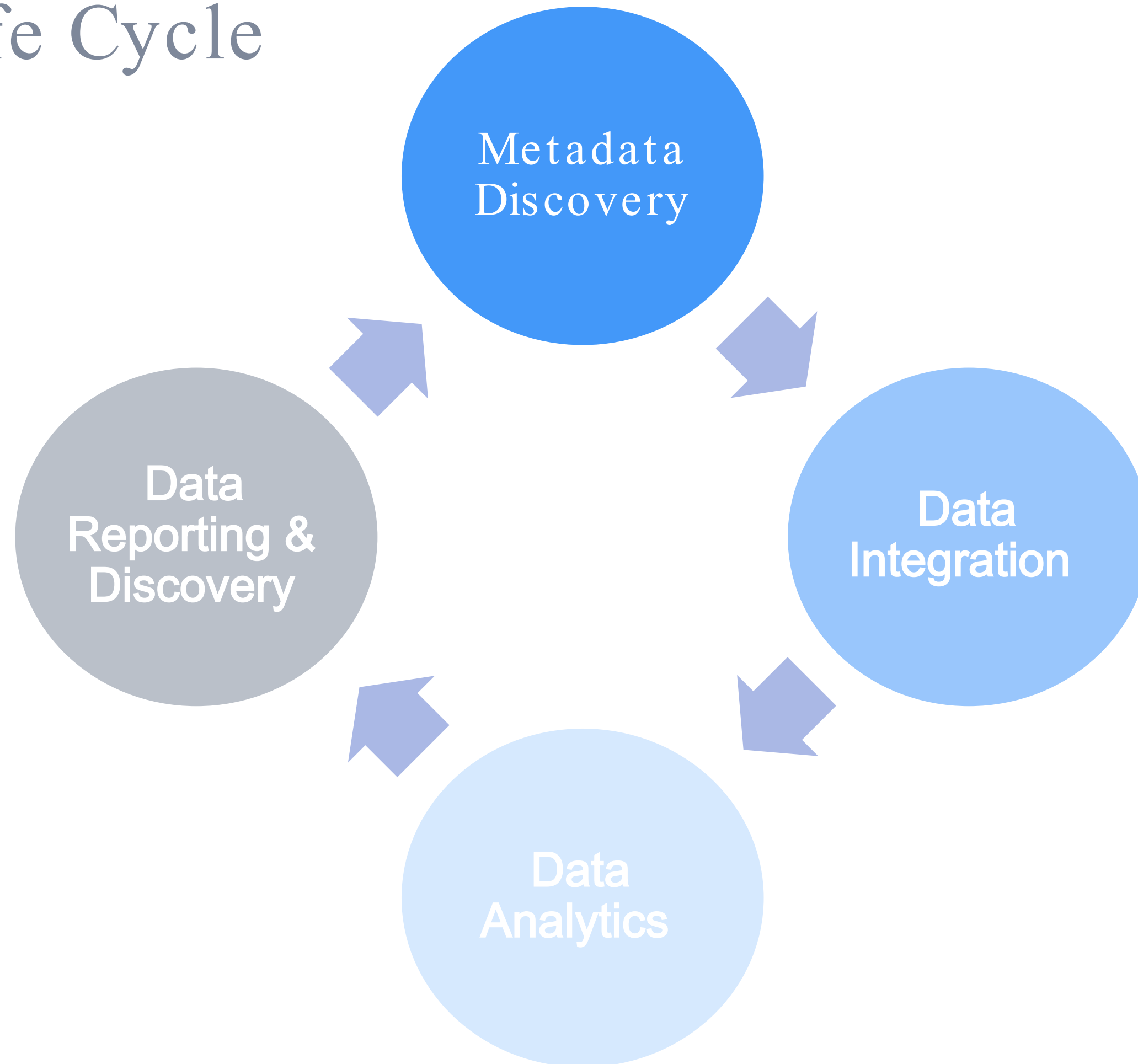
A SAS Instructor since 2007, I create content across the SAS YouTube channel, global forum, Ask the Expert series, and Training Post blog.

I teach at conferences on SAS, SQL, PERL, Macros, Python, Viya, and more.

Off the clock, I'm a yoga instructor exploring Canadian trails with my husky Miko.



Analytics Life Cycle



1 Metadata Discovery

SAS/ ACCESS LIBNAME Statement

SAS/ ACCESS has multiple LIBNAME engines that access Microsoft Excel workbooks.

Engine	Status
XLSX	 Current/ recommended for .xlsx files
EXCEL	Older Windows-based engine
PCFILES	Older option requiring PC Files Server in some scenarios
XLS	Legacy Excel 97-2003 format

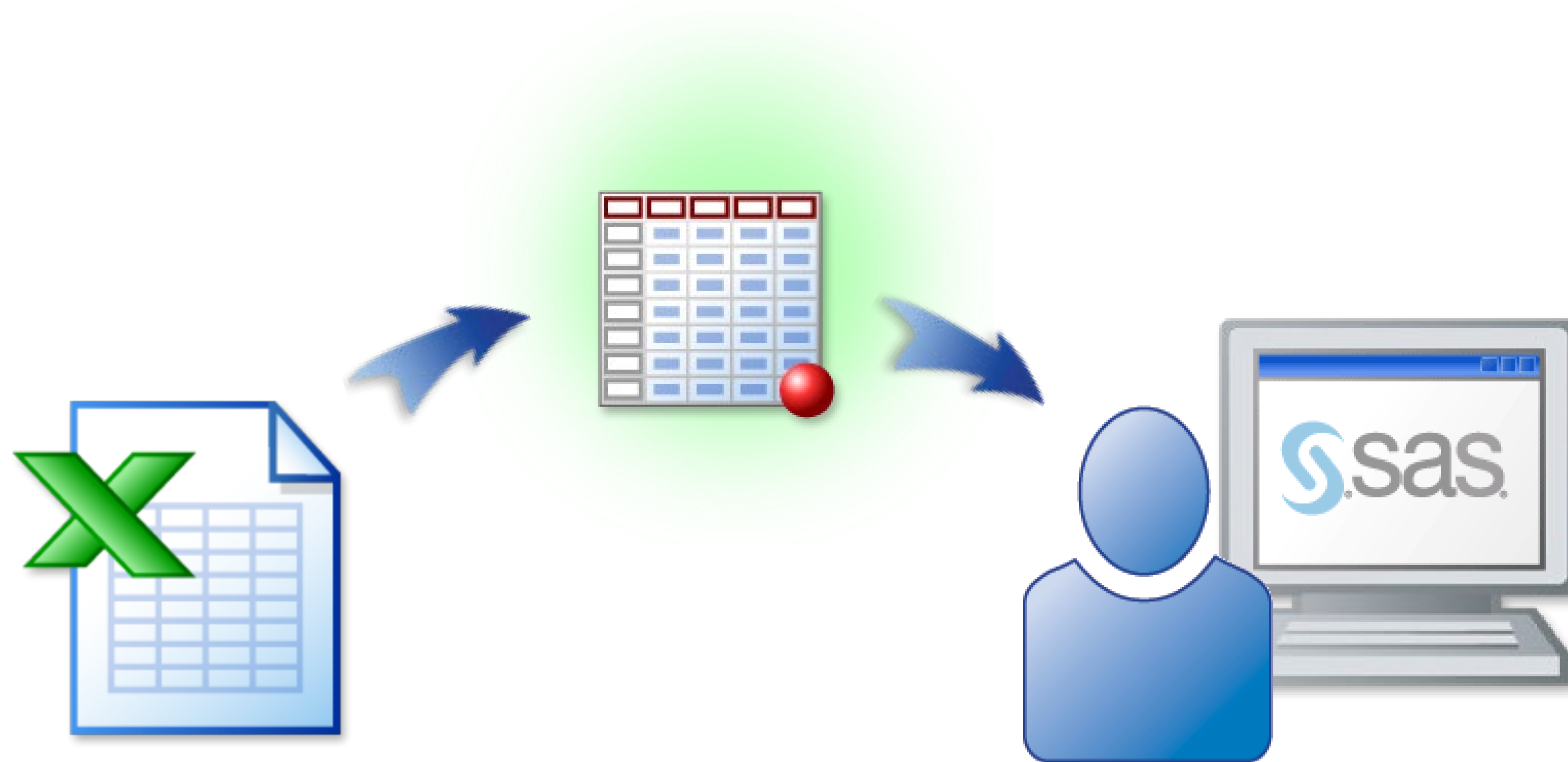
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Requirement	XLSX Engine
Microsoft Office installed?	 No
32-bit vs 64-bit Office match SAS?	 No
SAS/ACCESS to PC Files licensed?	 Yes

With the XLSX engine, SAS talks directly to the Excel workbook. Microsoft Office doesn't even need to be installed, so 32-bit versus 64-bit Office is no longer a concern.

Business Scenario

Use SAS/ ACCESS Interface to PC Files to read the Excel worksheet as if it were a SAS data set.



Examining Cars.xlsx

AutoSave On cars.xlsx Public • Saved

File Home Insert Page Layout Formulas Data Review View Automate Help Acrobat

Clipboard Font Alignment Number Styles Cells Editing Sensitivity Add-ins Adobe Acrobat

Comments Share

F1 MSRP

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Make	Model	Type	Origin	DriveTrain	MSRP	Invoice	EngineSize	Cylinders	Horsepower	MPG_City	MPG_Highway	Weight	Wheelbase	Length	ZIP
2	Acura	MDX	SUV	Asia	All	36945	33337	3.5	6	265	17	23	4451	106	189	
3	Acura	RSX Type S 2dr	Sedan	Asia	Front	23820	21761	2	4	200	24	31	2778	101	172	
4	Acura	TSX 4dr	Sedan	Asia	Front	26990	24647	2.4	4	200	22	29	3230	105	183	
5	Acura	TL 4dr	Sedan	Asia	Front	33195	30299	3.2	6	270	20	28	3575	108	186	
6	Acura	3.5 RL 4dr	Sedan	Asia	Front	43755	39014	3.5	6	225	18	24	3880	115	197	
7	Acura	3.5 RL w/Navigation 4dr	Sedan	Asia	Front	46100	41100	3.5	6	225	18	24	3893	115	197	
8	Acura	NSX coupe 2dr manual S	Sports	Asia	Rear	89765	79978	3.2	6	290	17	24	3153	100	174	
9	Audi	A4 1.8T 4dr	Sedan	Europe	Front	25940	23508	1.8	4	170	22	31	3252	104	179	
10	Audi	A4 1.8T convertible 2dr	Sedan	Europe	Front	35940	32506	1.8	4	170	23	30	3638	105	180	
11	Audi	A4 3.0 4dr	Sedan	Europe	Front	31840	28846	3	6	220	20	28	3462	104	179	
12	Audi	A4 3.0 Quattro 4dr manual	Sedan	Europe	All	33430	30366	3	6	220	17	26	3583	104	179	
13	Audi	A4 3.0 Quattro 4dr auto	Sedan	Europe	All	34480	31388	3	6	220	18	25	3627	104	179	
14	Audi	A6 3.0 4dr	Sedan	Europe	Front	36640	33129	3	6	220	20	27	3561	109	192	
15	Audi	A6 3.0 Quattro 4dr	Sedan	Europe	All	39640	35992	3	6	220	18	25	3880	109	192	
16	Audi	A4 3.0 convertible 2dr	Sedan	Europe	Front	42490	38325	3	6	220	20	27	3814	105	180	
17	Audi	A4 3.0 Quattro convertible 2dr	Sedan	Europe	All	44240	40075	3	6	220	18	25	4012	105	180	

carsdeal

Ready Accessibility: Good to go Display Settings 100%

one worksheet

XLSX Engine

```
LIBNAME libref XLSX "workbook-name.xlsx" <options>;
```

XLSX Engine Key Characteristics

- SAS 9.4M2+ and SAS Viya
- Requires SAS/ACCESS Interface to PC Files
- Works on Windows, Linux, and UNIX
- No Microsoft Office installation required
- No 32-bit / 64 -bit Office dependency
- Directly accesses .xlsx workbooks (and xlsxm read support in modern Viya)
- Supports large worksheets (>255 columns)
- Uses a simplified set of LIBNAME options
- No PC Files Server required

Exploring the Library

Regardless of the LIBNAME engine that is used, you can use the CONTENTS procedure to explore the library.

```
/* Point to Excel workbook */  
libname cars xlsx "&path\cars.xlsx";  
  
proc contents data=cars.carsdeal;  
run;  
proc contents data=sashelp.zipcode;  
run;
```

When SAS has a libref that is assigned to an Excel workbook, the workbook cannot be opened in Excel.

Exploring the Library

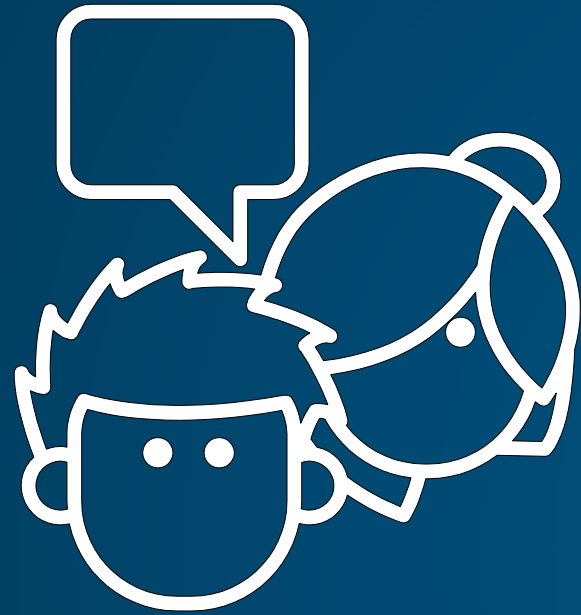
Data Set Name	CARS.carsdeal	Observations	.
Member Type	DATA	Variables	16
Engine	XLSX	Indexes	0
Created	.	Observation Length	0
Last Modified	.	Deleted Observations	0
Protection		Compressed	NO
Data Set Type		Sorted	NO
Label			
Data Representation	Default		
Encoding	Default		

Alphabetic List of Variables and Attributes						
#	Variable	Type	Len	Format	Informat	Label
9	Cylinders	Num	8	BEST.		Cylinders
5	DriveTrain	Char	5	\$5.	\$5.	DriveTrain
8	EngineSize	Num	8	BEST.		EngineSize
10	Horsepower	Num	8	BEST.		Horsepower
7	Invoice	Num	8	15.		Invoice
15	Length	Num	8	BEST.		Length
11	MPG_City	Num	8	BEST.		MPG_City
12	MPG_Highway	Num	8	BEST.		MPG_Highway
6	MSRP	Num	8	15.		MSRP
1	Make	Char	13	\$13.	\$13.	Make
2	Model	Char	40	\$40.	\$40.	Model
4	Origin	Char	6	\$6.	\$6.	Origin
3	Type	Char	6	\$6.	\$6.	Type
13	Weight	Num	8	BEST.		Weight
14	Wheelbase	Num	8	BEST.		Wheelbase
16	ZIP	Num	8	BEST.		ZIP

Data Set Name	SASHELP.ZIPCODE	Observations	41029
Member Type	DATA	Variables	21
Engine	V9	Indexes	3
Created	2023-01-18 20:50:56	Observation Length	888
Last Modified	2023-01-18 20:50:56	Deleted Observations	0
Protection		Compressed	NO
Data Set Type		Sorted	YES
Label	US Zipcodes; Source: zipcodedownload.com October 2022		
Data Representation	WINDOWS_64		
Encoding	us-ascii ASCII (ANSI)		

Alphabetic List of Variables and Attributes						
#	Variable	Type	Len	Format	Label	
18	ALIAS_CITY	Char	300		USPS - alternate names of city separated by	
19	ALIAS_CITYN	Char	300		Local - alternate names of city separated by	
12	AREACODE	Num	8		Single Area Code for ZIP Code.	
13	AREACODES	Char	12		Multiple Area Codes for ZIP Code.	
5	CITY	Char	35		Name of city/org	
20	CITY2	Char	35		Clean CITY name for geocoding	
9	COUNTY	Num	8		FIPS county code.	
10	COUNTYNM	Char	30		Name of county/parish.	
16	DST	Char	1		ZIP Code obeys Daylight Savings: Y-Yes N-No	
15	GMTOFFSET	Num	8		Diff (hrs) between GMT and time zone for ZIP Code	
11	MSA	Num	8		Metro Statistical Area code by common pop-pre 2003; no MSA for rural	
17	PONAME	Char	35		USPS Post Office Name: same as City	
6	STATE	Num	8		Two-digit number (FIPS code) for state/territory	
7	STATECODE	Char	2		Two-letter abbrev. for state name.	
8	STATENAME	Char	30		Full name of state/territory	
21	STATENAME2	Char	30		Clean STATENAME for geocoding	
14	TIMEZONE	Char	9		Time Zone for ZIP Code.	
2	X	Num	8	8.3	Longitude (degrees) of the center (centroid) of ZIP Code.	
3	Y	Num	8	8.3	Latitude (degrees) of the center (centroid) of ZIP Code.	
1	ZIP	Num	8	Z5.	The 5-digit ZIP Code	
4	ZIP_CLASS	Char	1		ZIP Code Classification: P=PO Box U=Unique zip used for large orgs/businesses/bldgs Blank=Standard/non-unique	

Quiz



What is the difference between using the `XLSX LIBNAME` engine and `PROC IMPORT` to read Excel data in a SAS program?

Exploring Common Columns

Dictionary Tables

```
title 'Identifying Common columns';

proc sql;
    select name, memname, libname
    from dictionary.columns
    where (libname='CARS' and memname='CARSDEAL')
        or (libname='SASHELP' and memname ne 'CARS')
    group by name
    having count(distinct libname)=2
    order by 1,3;

quit;
```

Identifying Common columns

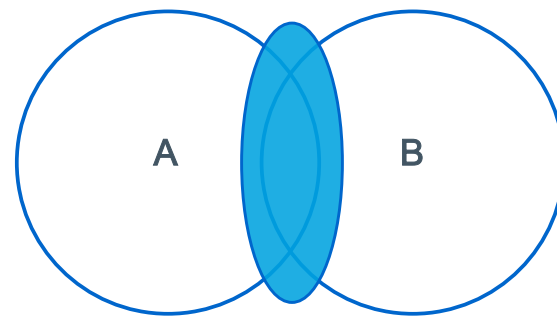
Column Name	Member Name	Library Name
Length	CARSDEAL	CARS
Length	LAKE	SASHELP
Length	COMET	SASHELP
Make	CARSDEAL	CARS
Make	JUNKMAIL	SASHELP
Type	CARSDEAL	CARS
Type	SVRTDIST	SASHELP
Type	QUAKES	SASHELP
Type	SLKWXL	SASHELP
Type	SPRINGS	SASHELP
Type	_CMPIDX_	SASHELP
Weight	CARSDEAL	CARS
Weight	FISH	SASHELP
Weight	HEART	SASHELP
Weight	BWEIGHT	SASHELP
Weight	GRIDDED	SASHELP
Weight	CLASS	SASHELP
Weight	CLASSFIT	SASHELP
ZIP	CARSDEAL	CARS
ZIP	ZIPMIL	SASHELP
ZIP	GEOEXM	SASHELP
ZIP	ZIPCODE	SASHELP

2 Data Integration

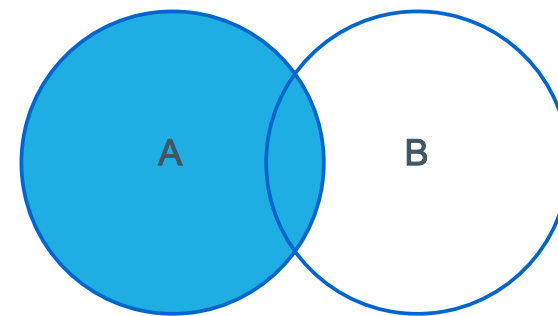
Data Integration

Investigate Common Columns

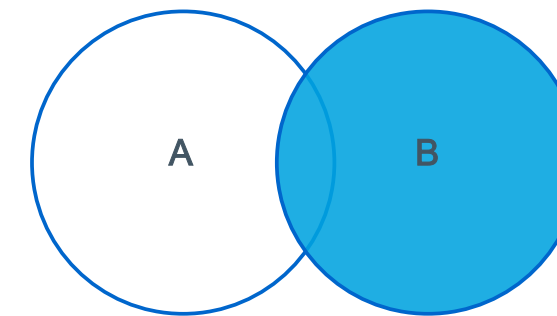
Use `DICTIONARY.COLUMNS` to find candidate join keys before combining Excel and SAS tables.



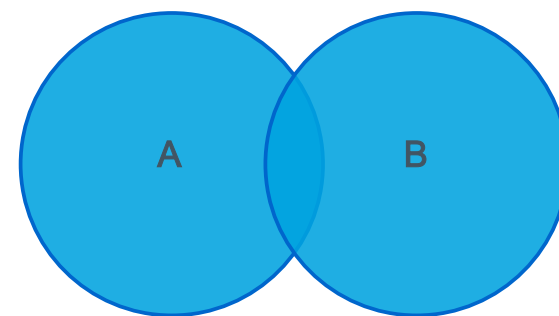
Inner Join



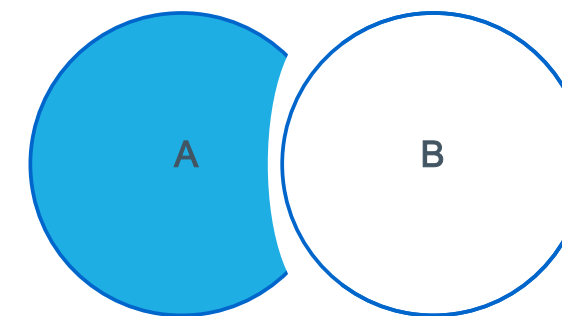
Left Outer Join



Right Outer Join



Full Outer Join



Minus / Left Only

Common column discovered → choose join type → preserve or filter matched rows

2 Data Integration

Joining Tables

🔥 Make	🔥 Model	🔥 Type	🔥 Origin	🔥 DriveTrain	🔗 MSRP	🔗 Invoice	🔗 EngineS
Toyota	Echo 2dr auto	Sedan	Asia	Front	11560	10896	
Subaru	Baja	Truck	Asia	All	24520	22304	
Jaguar	S-Type 3.0 4dr	Sedan	Europe	Rear	43895	40004	
Lexus	IS 300 4dr auto	Sedan	Asia	Rear	32415	28611	
Mitsubishi	Montero XLS	SUV	Asia	All	33112	30763	
Subaru	Impreza WRX 4dr	Sports	Asia	All	25045	23022	

```
/*2 Data Integration*/
```

```
proc sql;  
    create table cars_enriched as  
        select c.*,  
            z.city,  
            z.statecode,  
            z.statename  
        from cars.carsdeal as c  
        left join sashelp.zipcode as z  
            on c.zip=z.zip;  
quit;
```


3 Data Analytics

Geographic Analysis

```
/*3A. Geographic Analysis*/

title '# of vehicles by state after ZIP-
code lookup';
proc sql;
    select StateName,
           count(*) as Vehicles
    from cars_enriched
    where not missing(StateName)
    group by StateName
    order by Vehicles desc;
quit;
```

of vehicles by state after ZIP-code lookup

Full name of state/territory	Vehicles
California	16
Arizona	16
Massachusetts	15
Washington	15
North Carolina	15
Florida	14
New York	14
Illinois	14
Georgia	14
Texas	14

Vehicle Type Analysis

```
/*3B. Vehicle Type Analysis*/  
title "Vehicle Type Distribution and Average  
MSRP";  
proc sql;  
    select Type,  
           count(*) as Vehicles,  
           mean(MSRP) format=dollar12. as Avg_MSRP  
    from cars_enriched  
    group by Type  
    order by Avg_MSRP desc;  
quit;
```

Vehicle Type Distribution and Average MSRP

Type	Vehicles	Avg_MSRP
Sports	49	\$53,387
SUV	60	\$34,790
Sedan	262	\$29,774
Wagon	30	\$28,841
Truck	24	\$24,941
Hybrid	3	\$19,920

Margin Analysis

```
/*3C Margin Analysis*/  
title 'Profit Margin by Vehicle Type';  
proc sql;  
    select Type,  
           mean(MSRP-Invoice) format=dollar12. as Avg_Margin,  
           max(MSRP-Invoice) format=dollar12. as Max_Margin  
    from cars_enriched  
    group by Type  
    order by Avg_Margin desc;  
quit;
```

Profit Margin by Vehicle Type

Type	Avg_Margin	Max_Margin
Sports	\$4,914	\$18,905
SUV	\$3,165	\$8,345
Sedan	\$2,384	\$8,820
Truck	\$2,325	\$4,941
Wagon	\$2,195	\$4,644
Hybrid	\$1,491	\$1,689

Luxury Segmentation

```
/*3D Luxury Segmentation*/  
title 'Luxury Vehicle Segmentation';  
  
proc sql;  
    select case  
        when MSRP > 50000 then 'Luxury'  
        else 'Standard'  
    end as Luxury_Segment,  
    count(*) as Vehicles,  
    mean(MSRP) format=dollar12. as Avg_MSRP,  
    mean(MSRP-Invoice) format=dollar12. as Avg_Margin  
from cars_enriched  
group by Luxury_Segment  
order by Avg_MSRP desc;  
quit;
```

Luxury Vehicle Segmentation

Luxury_Segment	Vehicles	Avg_MSRP	Avg_Margin
Luxury	52	\$72,595	\$6,508
Standard	376	\$27,268	\$2,242

4 Reporting & Delivery

4 Reporting & Delivery

```
title 'Excel-Based Vehicle Analytics Report';  
ods excel file="c:\temp\cars_report.xlsx";  
  
ods excel options(sheet_name="State Summary");  
  
Embed all code here  
ods excel close;
```

State Summary

Type Summary

Margin Analysis

Luxury Segmentation

Handy Links

- [Github link for all files used in the presentation and demo](#)
- [LIBNAME Statement: XLSX Engine \(SAS Help Center\)](#)
- [Using LIBNAME XLSX to Read and Write Excel Files \(Chris Hemedinger\)](#) – Explains the motivation behind the XLSX engine and compares it with EXCEL and PCFILES engines.
- [Using LIBNAME XLSX to Read and Write Excel Files \(9to5SAS\)](#) – Practical examples for reading, writing, and navigating Excel workbooks using the XLSX engine. [I](#)
- [Tips to Read In and Output Excel Spreadsheets in SAS® \(PharmaSUG 2023\)](#) – Comprehensive paper covering Excel formats, import/export methods, and SAS integration techniques.
- [SAS/ACCESS for PC Files Documentation](#) – Reference documentation covering SAS/ACCESS Interface to PC Files, licensing considerations, and Excel connectivity options.

Thank You

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✓ Did you
enjoy this
session, Let us
know in the
[evaluation](#)

